Lecture 24: Periodic Motion

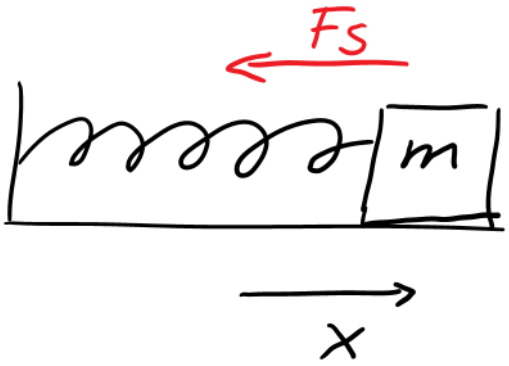
- Motion of a mass at the end of a spring
- Differential equation for simple harmonic oscillation
- Amplitude, period, frequency and angular frequency
- Energetics
- Simple pendulum
- Physical pendulum

Mass at the end of a spring

Mass m connected to a spring with spring constant k on a frictionless surface

$$F_x = -kx$$

Linear restoring spring force



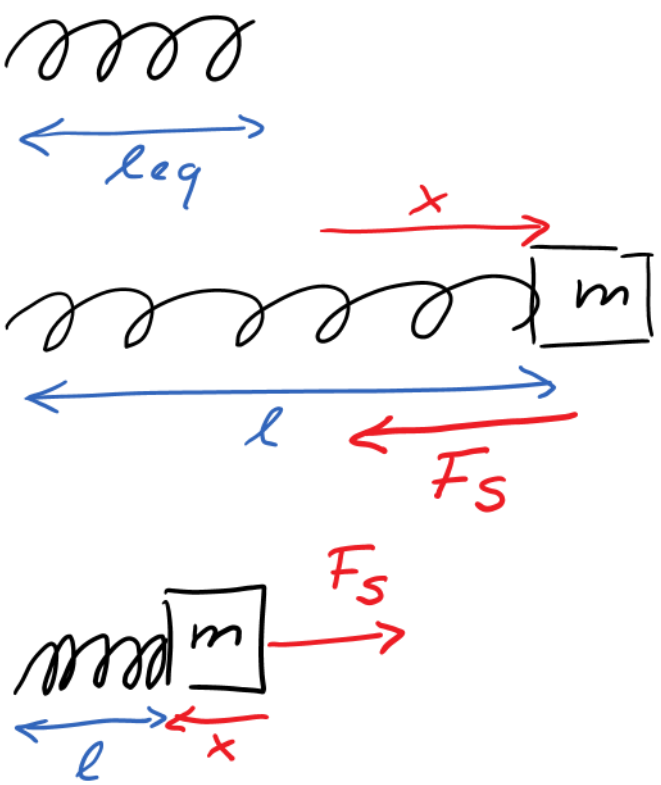
Spring force

Spring force: $F_{S\,x} = -kx$

$x = l - l_{eq}$

stretch or compression

k force constant



F_x is negative if x is positive (stretched spring)

F_x is positive if x is negative (compressed spring)

Differential equation of a SHO

Newton's 2nd Law: $\sum F_x = ma_x$

$$-kx = m\frac{d^2x}{dt^2}$$

$$-\frac{k}{m}x = \frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} = -\omega^2 x \quad *$$

$$\omega = \sqrt{\frac{k}{m}}$$ Angular frequency

Differential equation of a Simple Harmonic Oscillator

*We can always write it like this because m and k are positive

Solution

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

Equation for SHO

General solution: $x = A\cos(\omega t + \varphi)$

Take 2nd derivative: $\frac{d^2x}{dt^2} = -\mathrm{A}\,\omega^2\cos(\omega t + \varphi) = -\omega^2 x$

A and φ: two "constants of integration" from solution of a *second-order* differential equation.
Determined by the initial conditions.

Amplitude

$$x = A\cos(\omega t + \varphi)$$

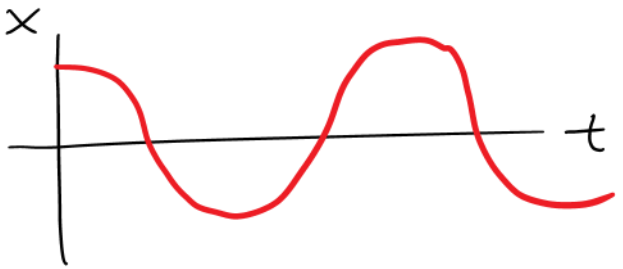
Range of cosine function: -1…+1

$\Longrightarrow \quad -A \leq x(t) \leq +A$

A = Amplitude of the oscillation

Phase Constant

$$x = A\cos(\omega t + \varphi)$$



If φ=0:

$x = A\cos(\omega t)$

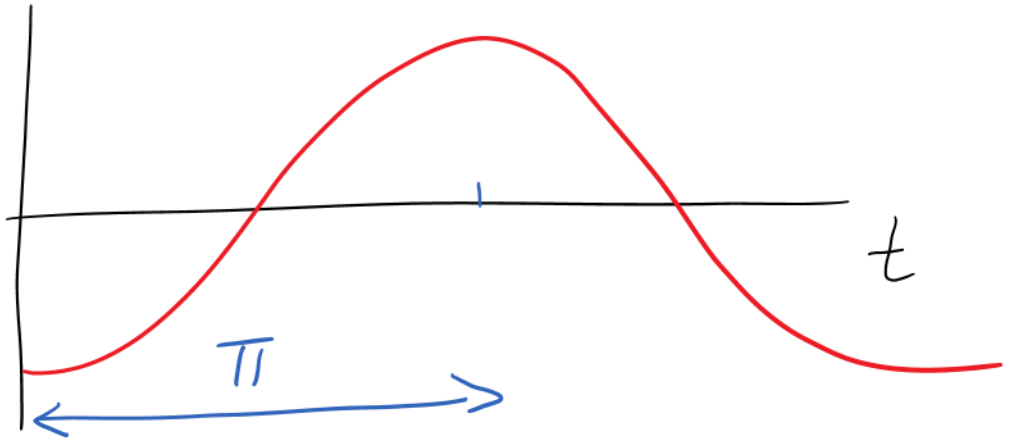
$x(t = 0) = x_0 = A$

To describe motion with different starting points:
Add phase constant to shift the cosine function

$$x = A\cos(\omega t + \varphi)$$

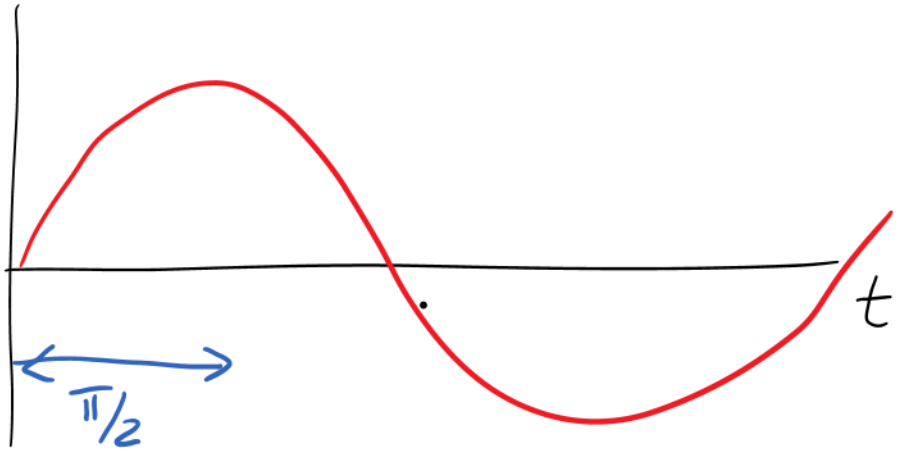
$X_0 = -A$:

shift by π



$X_0 = 0$:

shift by $\frac{\pi}{2}$



Initial conditions

$$x_0 = x(t = 0)$$
$$v_{x0} = v_x(t = 0)$$

$$x_0 = A\cos(0 + \varphi) = A\cos(\varphi)$$
$$v_{x0} = -A\omega\, sin(0 + \varphi) = -A\omega\, sin(\varphi)$$

$\rightarrow$ two equations for A and φ

Position and velocity

$$x = A\cos(\omega t + \varphi)$$

$$v_x = \frac{dx}{dt} = -A\omega \sin(\omega t + \varphi)$$

At time t_m: $x = x_{max} = A$ $\cos(\omega t_m + \varphi) = 1$

$(\omega t_m + \varphi) = 0 \; or \; \pi$

$\sin(\omega t_m + \varphi) = 0$ $\Rightarrow$ $v_x(t_m) = 0$

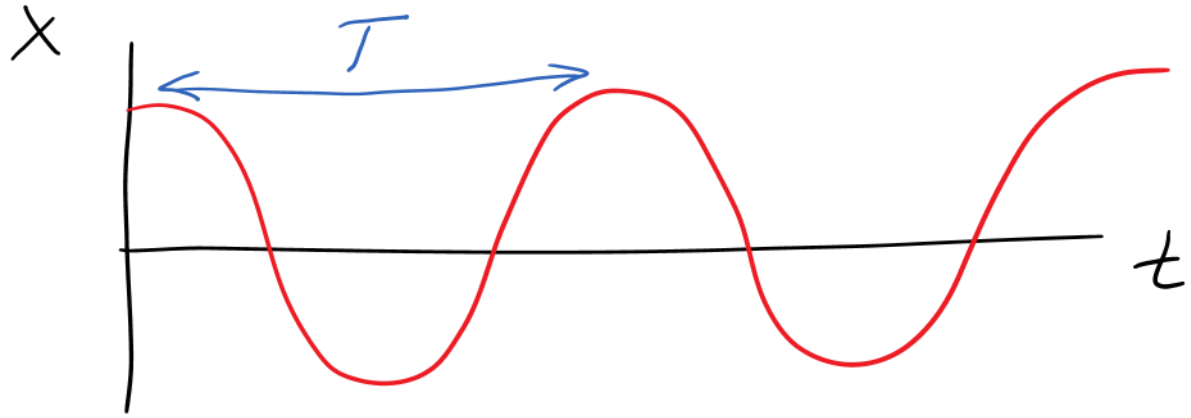
Mass stops and reverses direction when it reaches maximum displacement (turning point)

Simulation

Walter Fendt Spring Pendulum Simulation

Period and angular frequency

$$x = A\cos(\omega t + \varphi)$$



Time T for one complete cycle: period

$(\omega t + \varphi)$ changes by 2π in time T

$$\omega T = 2\pi \quad \Longrightarrow \quad \omega = \frac{2\pi}{T} = 2\pi f$$

Effect of mass and amplitude on period

$$\omega T = 2\pi \implies T = \frac{2\pi}{\omega}$$

$$\omega = \sqrt{\frac{k}{m}} \qquad \implies T = \frac{2\pi}{\sqrt{\frac{k}{m}}}$$

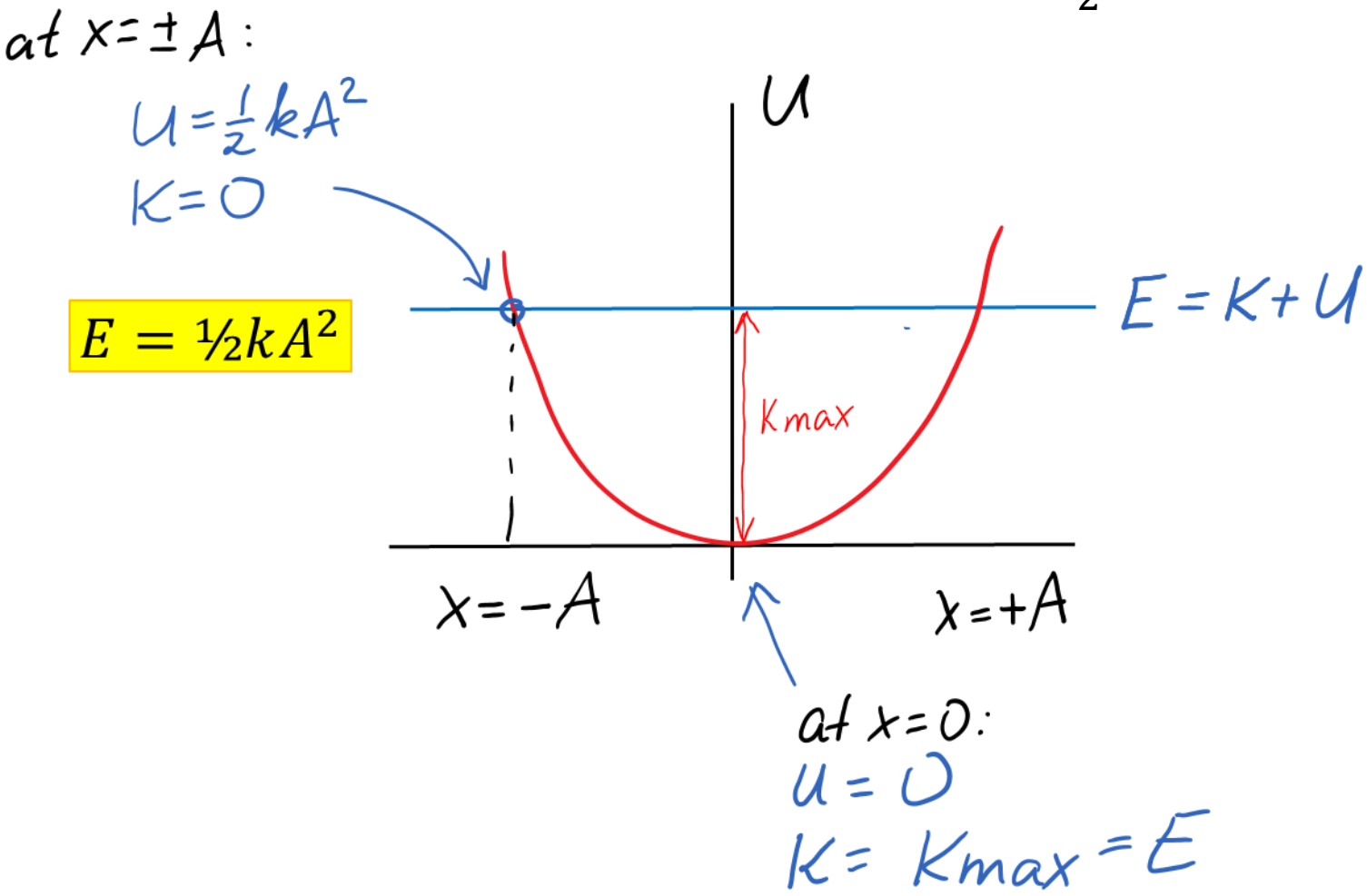
$$T = 2\pi\sqrt{\frac{m}{k}}$$

Amplitude A does not appear – no effect on period

Demo: Vertical springs showing effect of m and A

Energy in SHO

Potential energy of spring force: $U = \frac{1}{2}kx^2$



Example

A block of mass *M* is attached to a spring and executes simple harmonic motion of amplitude *A*. At what displacement(s) *x* from equilibrium does its kinetic energy equal twice its potential energy?

General SHO

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

Equation for SHO

General solution:

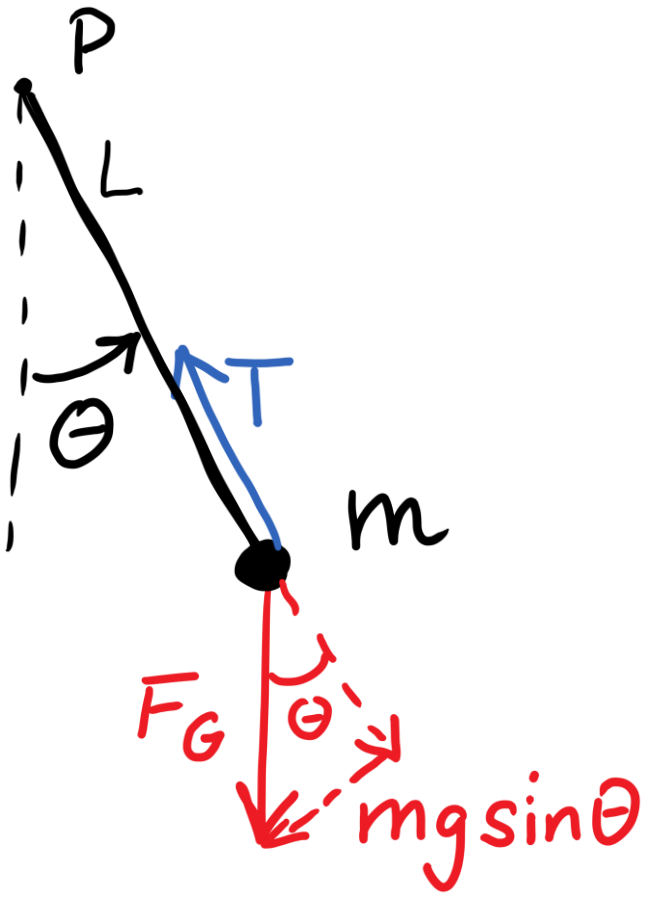
$$x = A\cos(\omega t + \varphi)$$

$$T = \frac{2\pi}{\omega}$$

Simple Pendulum

Point mass m at the end of a massless string of length L

θ = displacement coordinate (**with sign**) from vertical equilibrium position



Simple Pendulum Oscillation

$$\Sigma\tau_z = I\alpha_z$$

$$-mg\ L\ sin\theta = mL^2\frac{d^2\theta}{dt^2}$$

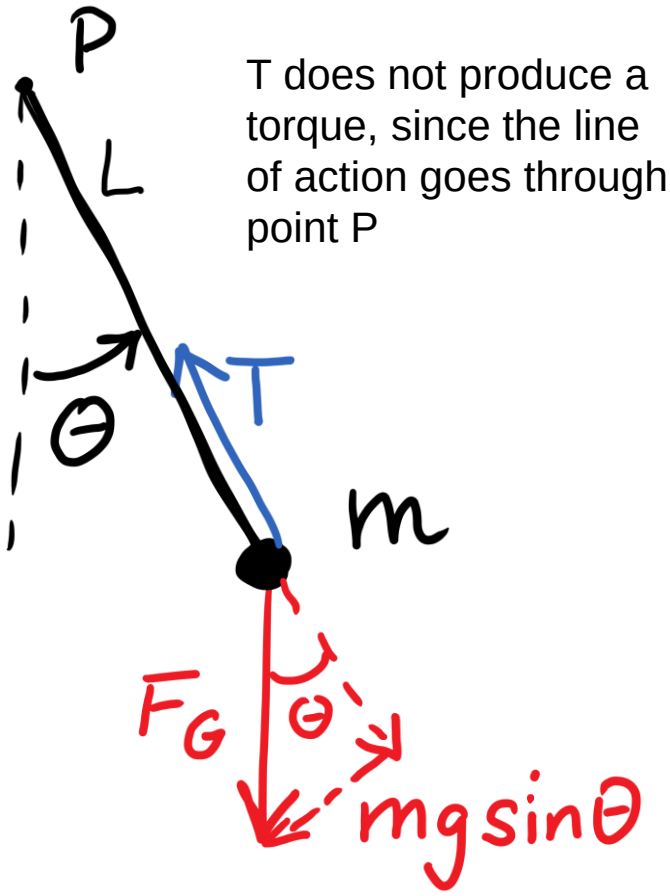
Very complicated differential equation!
But for small oscillations:

$$sin\theta \approx \theta$$

And

$$-\frac{g}{L}\theta = \frac{d^2\theta}{dt^2}$$

Differential equation of SHO



T does not produce a torque, since the line of action goes through point P

Simple Pendulum Period

$$-\frac{g}{L}\theta = \frac{d^2\theta}{dt^2}$$

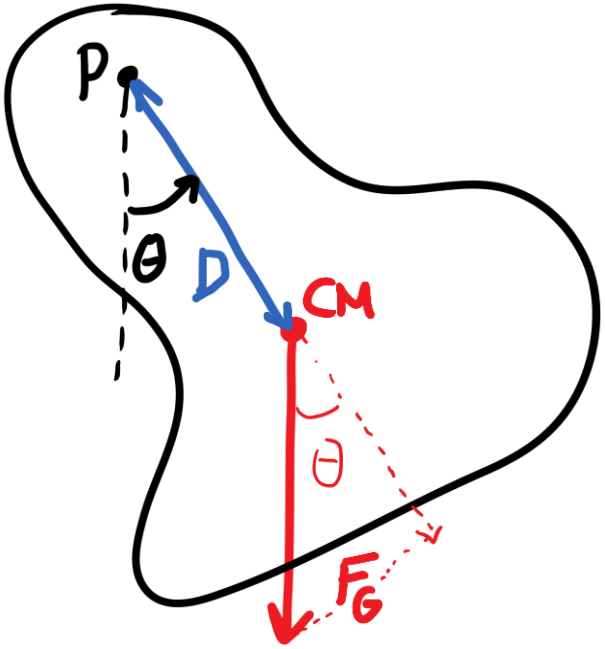
Differential equation of simple harmonic oscillator

$$\theta(t) = \theta_{max}\cos(\omega t + \varphi)$$

With $\omega = \sqrt{\frac{g}{L}}$ and $T = 2\pi\sqrt{\frac{L}{g}}$

- Period independent of mass
- Period independent of amplitude

Physical Pendulum



Extended object of mass m that swings back and forth about an axis ***P*** that does not go through its center of mass **CM**.

$$\Sigma\tau_z = I\alpha_z$$

$$-mg\ D\ sin\theta = I\frac{d^2\theta}{dt^2}$$

For small oscillations: $sin\theta \approx \theta$

$$-\frac{mgD}{I}\theta = \frac{d^2\theta}{dt^2}$$

Physical Pendulum Oscillation

$$-\frac{mgD}{I}\theta = \frac{d^2\theta}{dt^2}$$

SHO:

$$\theta(t) = \theta_{max}\cos(\omega t + \varphi)$$

$$\omega = \sqrt{\frac{mgD}{I}}$$

$$T = 2\pi\sqrt{\frac{I}{mgD}}$$

I is moment of inertia about axis P

D is distance between P and CM

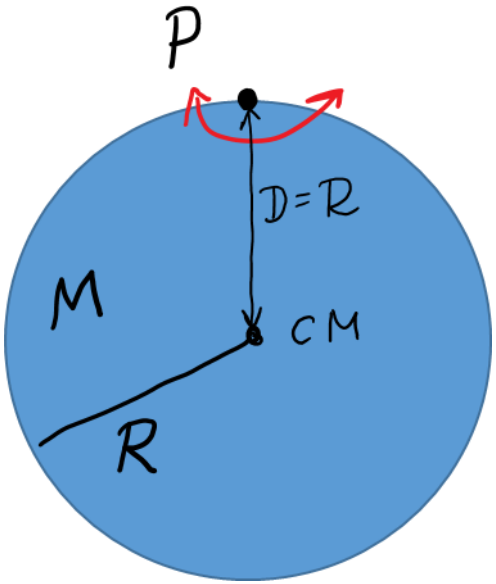
Parallel axis theorem:

$$I_P = I_{CM} + mD^2$$

Demo: Meter stick pivoted at different positions

Physicsl Pendulum Example

A uniform disk of mass M and radius R is pivoted at a point at the rim. Find the period for small oscillations.



$$T = 2\pi\sqrt{\frac{I}{mgD}}$$

$$I_P = I_{CM} + mD^2$$